

MANUFACTURING GOVERNANCE

Quality, OEE & Predictive Maintenance Control Tower

Detailed governance report for a deterministic synthetic factory, time-aware predictive maintenance model and human-gated decision policy.

SCOPE BOUNDARY

Recommendation only. No automated work orders, PLC/SCADA writes, equipment stops or maintenance approvals.

OEE	FPY	OOT PR-AUC	OOT RECALL
85.27%	95.27%	0.389	73.51%

Contents and evidence chain

The report follows one governed evidence chain: synthetic events become auditable KPIs, time-split predictions, explanations and planner-reviewed recommendations.

Section	Evidence topic	Section	Evidence topic
01	Executive decision record	11	Champion/challenger validation
02	Scope and claim boundary	12	Calibration and alarm quality
03	Deterministic data design	13	SHAP explanations
04	Data contract and quality	14	Maintenance decision policy
05	System architecture	15	Service, database and BI
06	OEE performance	16	Security and threat model
07	Downtime and reliability	17	Monitoring and incident response
08	Quality economics and capability	18	Risk register and limitations
09	SPC method and alarms	19	Reference crosswalk
10	Temporal model development	20	Delivery matrix

REVIEW PRINCIPLE

Metrics in this report are read directly from pipeline artifacts. No model or business result is manually fabricated.

- Primary evidence: artifacts/results/pipeline_metrics.json and exported analytical CSVs.
- Reproducibility: seed 20260729, documented configuration and executable pipeline.
- Communication: the same results feed README, Excel, PBIP, PowerPoint and this report.
- Reference mapping: official sources are design inputs, never certification claims.

The plant clears its OEE target; alert workload is the open control

OEE 85.27%	AVAILABILITY 98.36%	PERFORMANCE 90.99%	QUALITY / FPY 95.27%
UNPLANNED STOP 2,466 h	COPQ 12,936,906	MTBF 145.96 h	MTTR 1.73 h

Decision area	Evidence	Disposition	Owner gate
OEE	85.27%; performance 90.99%	Investigate speed loss before availability	Operations
Quality	FPY 95.27%; PRD-A Ppk 0.92	Capability investigation	Quality engineering
Model	OOT PR-AUC 0.389; recall 73.51%	Useful ranking; capacity watch	Model owner
Maintenance	5 P1 recommendations	Planner review only	Maintenance planner

OPEN OPERATIONAL CONTROL

OOT alert rate is 43.11%, above the 40% validation review-capacity cap. The report treats this as a watch item instead of suppressing it.

- All results describe synthetic data and synthetic cost units.
- The system can rank and explain; it cannot execute maintenance.
- Real-plant adoption would require shadow validation, plant-specific cost calibration and safety ownership.

What the platform does—and explicitly does not do

In scope	Out of scope
Compute OEE, downtime, reliability, quality, SPC and capability metrics.	Control production equipment or safety systems.
Estimate 24-hour failure probability and surface SHAP reason codes.	Create, approve or close CMMS work orders.
Combine risk, criticality and synthetic economics into planner priorities.	Claim real-world avoided cost or maintenance ROI.
Serve recommendations through a read-only FastAPI contract.	Write to PLC, SCADA, historian or enterprise maintenance systems.
Map design evidence to official references.	Claim certification, conformance or regulatory compliance.

HUMAN APPROVAL INVARIANT

Every response includes `human_approval_required=true` and `execution_mode=recommendation_only`. The decision boundary is enforced in code, contract tests and documentation.

CLAIM LANGUAGE

Approved interpretation

- Use: “synthetic demonstration,” “decision support,” “estimated probability,” and “proposed pilot.”
- Avoid: “production deployed,” “automatically schedules maintenance,” “prevents failures,” or “compliant with ISO/NIST.”
- Treat costs as scenario inputs; do not present them as financial statements.
- Treat SHAP values as model attribution, not root-cause proof.

Control	Evidence
API contract	<code>src/manufacturing_ct/api.py</code> and <code>tests/test_api.py</code>
Architecture decision	<code>docs/adr/001-recommendation-only-boundary.md</code>
Threat mitigation	<code>docs/governance/threat_model.md</code>
Runtime policy	<code>artifacts/results/policy_config.json</code>

A synthetic plant with realistic operating structure

LINES 4	MACHINES 12	PRODUCTS 4	SHIFTS 3
MACHINE SHIFTS 19,692	CTQ READINGS 98,460	DOWNTIME EVENTS 4,836	MAINTENANCE EVENTS 1,087

Entity	Grain	Examples	Purpose
Production shift	machine x 8-hour shift	counts, speed, sensors, lags	OEE and model features
Quality measurement	five CTQ readings / subgroup	dimension, specs, defect counts	SPC and capability
Downtime event	event	planned flag, failure type, duration	Pareto and reliability
Maintenance event	event	action, hours, synthetic cost	history and policy
Prediction	pre-shift decision record	probability, threshold, explanations	human review

REPRODUCIBILITY

Seed 20260729 • period 2024-01-01 to 2025-06-30 • the same configuration reproduces the exported metrics and figures.

- Multiple failure modes: electrical, bearing, sensor, hydraulic and overheating.
- Planned/unplanned downtime remains distinct from quality and maintenance records.
- Sensor degradation, product mix, seasonality and maintenance history generate learnable—not perfectly separable—risk.
- No real operator, equipment or plant data is present.

Contracts define grain, availability time and failure behavior

Contract control	Implementation	Failure disposition
Primary key uniqueness	shift_id at machine-shift grain	Fail pipeline
Required field completeness	pre-feature contract checks	Fail or quarantine
Non-negative counts	units, downtime and repair duration	Fail pipeline
Downtime within shift	event duration bounded by shift	Fail pipeline
Sensor plausibility	documented synthetic ranges	Flag and investigate
Schema/version	data contract + deterministic config	Block incompatible change

Pipeline check	Result	Exceptions / context
primary_key_unique	PASS	0
required_fields_complete	PASS	16682
non_negative_counts	PASS	0
downtime_within_shift	PASS	0
sensor_ranges_plausible	PASS	0

QUALITY RESULT

5/5 gates passed. Completeness context is recorded rather than silently discarded.

- Temporal availability is part of the contract: model features must exist before the shift decision timestamp.
- Prediction labels and realized outcomes are stored separately to preserve auditability.
- Data quality is not model drift; each has distinct thresholds and owners.

Five analytical layers with one recommendation-only boundary

Layer	Component	Responsibility	Evidence
1	Synthetic generator	events, sensors, quality and maintenance history	versioned CSV/artifacts
2	KPI + SPC analytics	OEE, Pareto, MTBF/MTTR, capability and alarms	analytical tables/figures
3	Temporal ML	fit, calibration, validation, OOT and threshold controls	model bundle/metrics
4	SHAP + policy	global/local attribution and cost-risk prioritization	reason codes/recommendations
5	Delivery	FastAPI, PostgreSQL views, PBIP, Excel, reports	human-facing outputs

GOVERNED ANALYTICAL RECORD

The same versioned metrics, model threshold and decision record feed every channel. This reduces narrative/model divergence across code, BI and executive reporting.

DEPLOYMENT PROFILES

Runtime separation

Profile	Purpose	Key controls
Docker Compose	local integrated stack	API, PostgreSQL, Prometheus and Grafana
Kubernetes	portable deployment starter	non-root, read-only filesystem, HPA, PDB, NetworkPolicy
PostgreSQL	governed data serving	normalized tables and analytical views
Prometheus/Grafana	operational observability	service/model metrics, alerts and dashboards

- No plant-network segmentation, identity-provider integration or enterprise secrets system is implemented.
- Production credentials must be externalized; the repository contains examples only.

Performance loss is the largest mathematical constraint

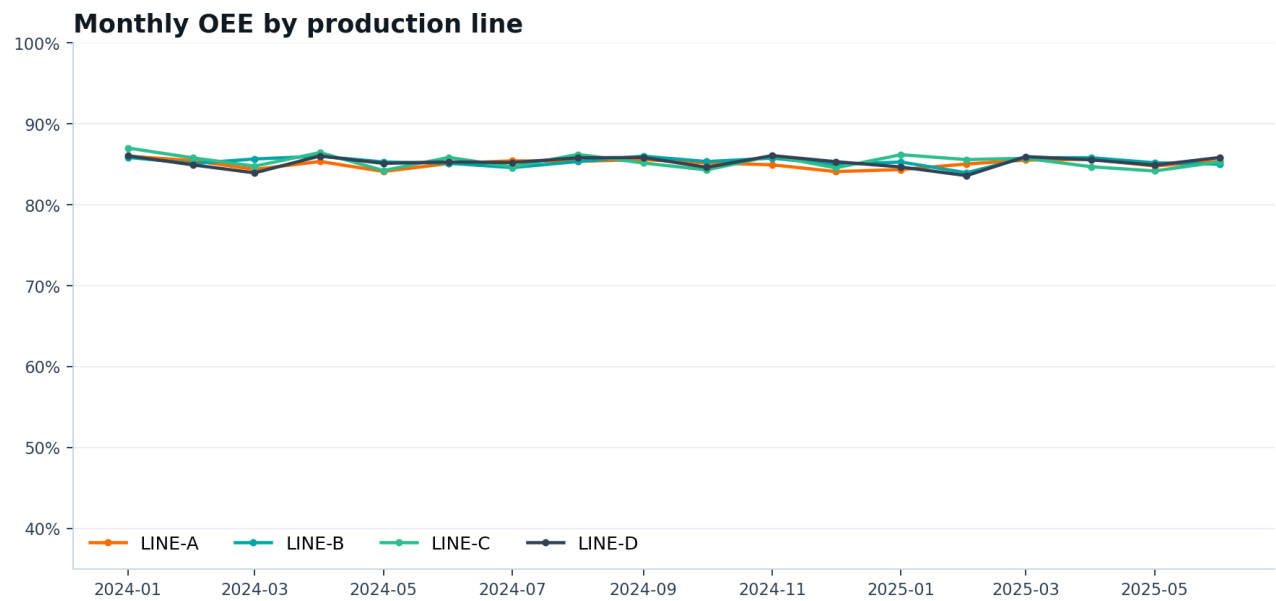


Figure 1. Monthly OEE by production line from deterministic pipeline output.

AVAILABILITY	PERFORMANCE	QUALITY	OEE
98.36%	90.99%	95.27%	85.27%

Component	Formula	Result	Interpretation
Availability	runtime / planned production time	98.36%	limited downtime loss
Performance	ideal output / actual runtime	90.99%	largest gap to 100%
Quality	first-pass good / total units	95.27%	quality opportunity remains
OEE	availability x performance x quality	85.27%	ratio-of-sums plant result

CALCULATION CONTROL

OEE uses runtime-weighted ratios and plant-level ratio-of-sums. Averaging row-level OEE would create weighting error.

Electrical events and minor stops explain most downtime

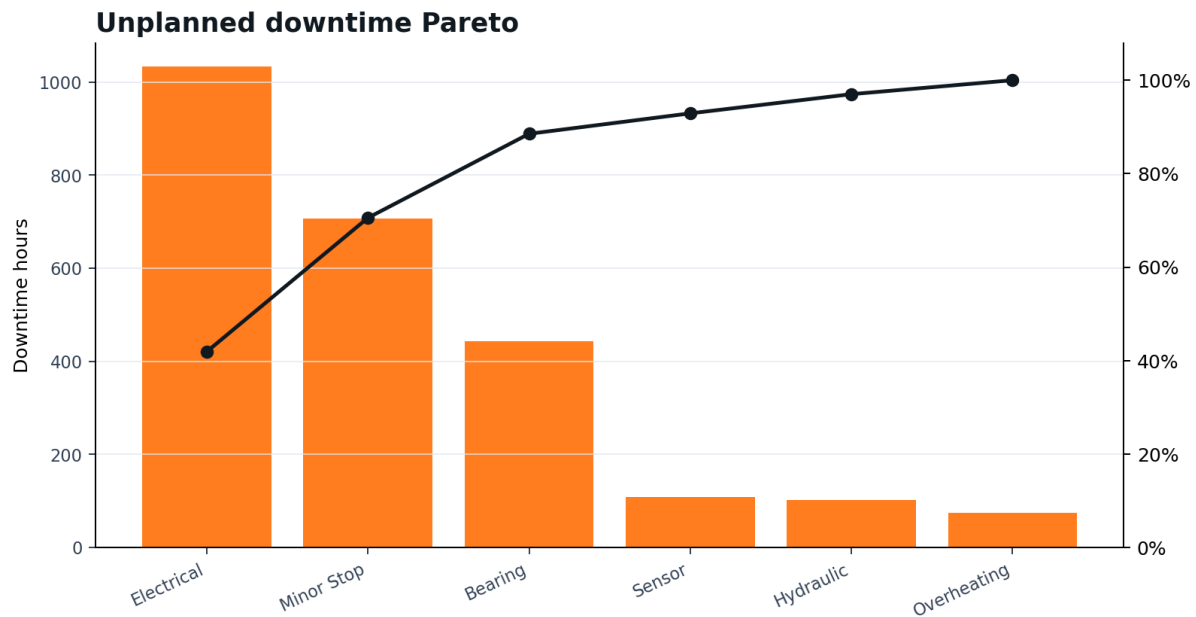


Figure 2. Unplanned downtime duration and cumulative share.

UNPLANNED HOURS	FAILURES	MTBF	MTTR
2,466	1,015	145.96 h	1.73 h

Category	Hours	Events	Share	Cumulative
Electrical	1,032.8	572	41.89%	41.89%
Minor Stop	706.4	3821	28.65%	70.53%
Bearing	443.8	199	18.00%	88.53%
Sensor	107.8	144	4.37%	92.90%
Hydraulic	101.3	51	4.11%	97.01%
Overheating	73.6	49	2.99%	100.00%

PRIORITY

Electrical and minor-stop countermeasures should precede lower-frequency categories. Reliability metrics describe history; they do not authorize maintenance.

PRD-A is the long-term capability watch item

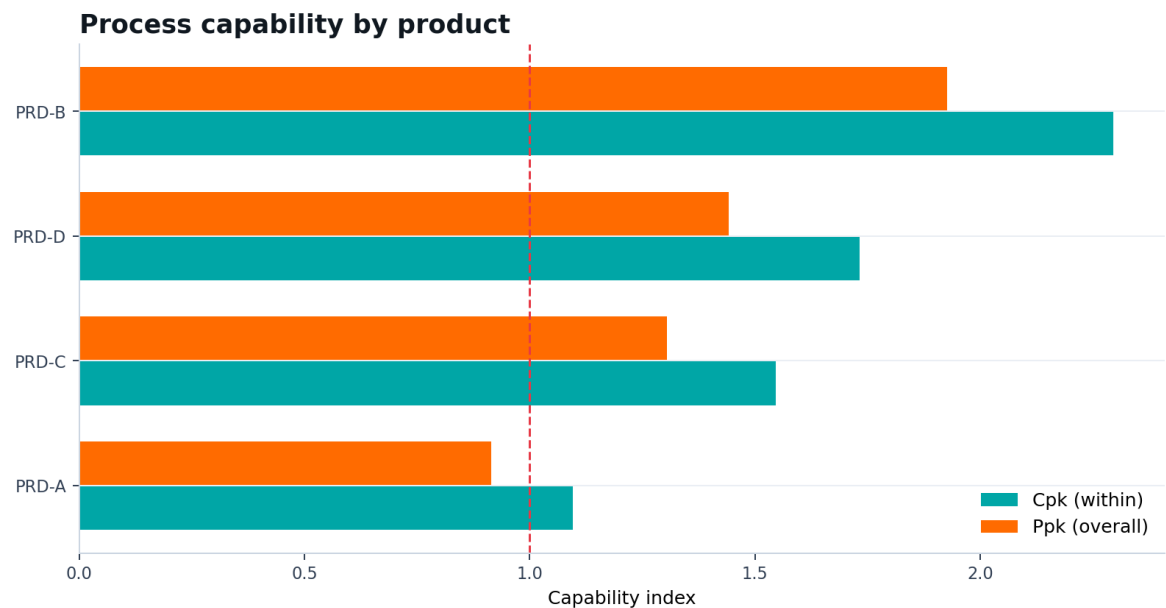


Figure 3. Within and overall capability indices by product.

FPY	SCRAP	REWORK	COPQ
95.27%	1.99%	2.75%	12,936,906

Product	Cp	Cpk	Pp	Ppk	Disposition
PRD-A	1.376	1.096	1.148	0.915	Watch long-term centering
PRD-B	2.575	2.295	2.161	1.927	Synthetic snapshot
PRD-C	1.822	1.546	1.537	1.304	Synthetic snapshot
PRD-D	2.015	1.733	1.677	1.442	Synthetic snapshot

INTERPRETATION

Cp/Cpk use within-subgroup variation; Pp/Ppk use overall variation. These are synthetic capability estimates—not product acceptance decisions.

Chart choice follows data type and sampling design

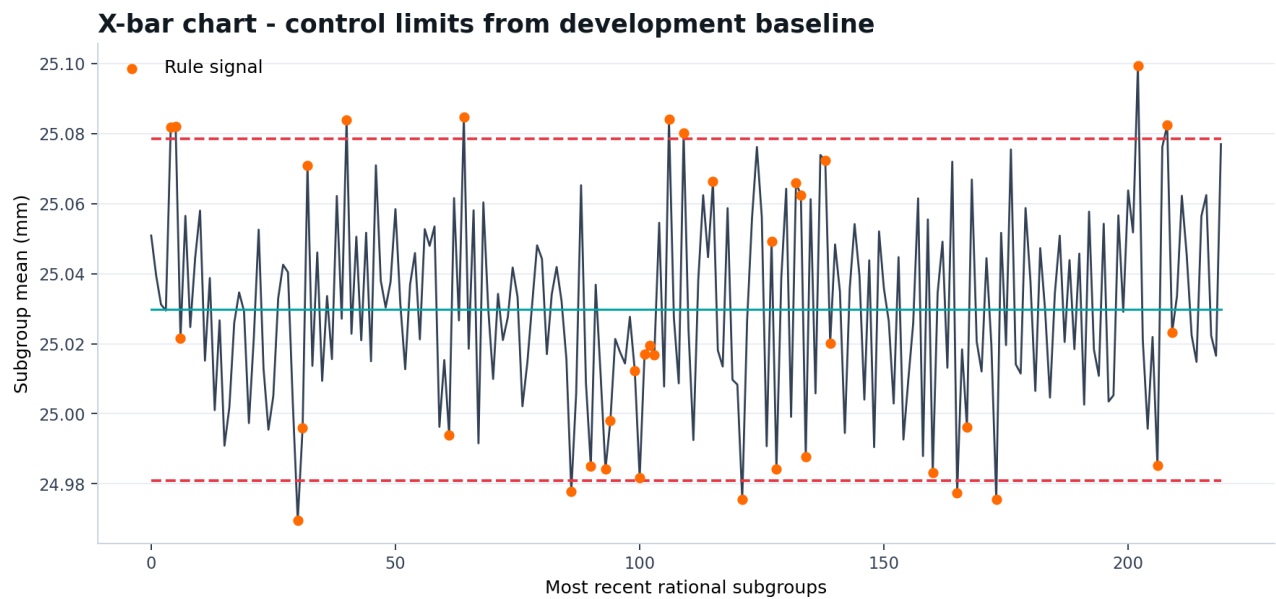


Figure 4. X-bar chart for fixed n=5 CTQ subgroups with rule signals.

Chart	Data	Sample-size condition	Why selected
X-bar/R	CTQ dimension	fixed n=5	subgroup mean and short-term range
I-MR	roughness	one observation / shift	individual level and moving range
p	defective proportion	variable lot size	binomial proportion limits vary by n
u	defects per unit	variable inspected units	Poisson rate limits vary by exposure
np / c	not selected	constant size required	assumption does not hold

Alarm rule	Documented trigger
Rule 1	one point beyond 3 sigma
Rule 2	two of three consecutive points beyond 2 sigma on one side
Rule 3	four of five consecutive points beyond 1 sigma on one side
Rule 4	eight consecutive points on one side of the center line

CRITICAL DISTINCTION

Control limits estimate baseline process behavior. LSL/USL are independent engineering specifications used for capability.

Model fit, calibration, validation and OOT remain chronologically isolated

Partition	Window	Rows	Positive rate	Allowed decisions
Calibration	2024-10-01 to 2024-11-29	2,160	15.69%	fit sigmoid calibration
Model Fit	2024-01-01 to 2024-09-29	9,828	15.35%	fit candidate models
Oot	2025-03-01 to 2025-06-29	4,356	15.43%	evaluate only
Validation	2024-12-01 to 2025-02-27	3,204	14.95%	select champion and threshold

LEAKAGE CONTROLS

Lagged and rolling features use past-only values; the 24-hour label looks forward. Chronological boundaries and purge gaps prevent observations from seeing future outcomes.

CLASS IMBALANCE AND THRESHOLDING

Decision controls

- Overall 24-hour positive rate: 15.29%.
- PR-AUC is compared with the positive-rate no-skill baseline, not interpreted in isolation.
- Threshold 0.1478 is chosen only on validation data.
- Validation constraints: recall target at least 72%; review-capacity cap 40%.
- OOT data does not alter model choice, calibration or threshold.

Control	Evidence
Feature availability	pre-shift feature construction and data contract
Chronological split	model_fit → calibration → validation → OOT
Calibration isolation	dedicated calibration partition
Threshold isolation	validation-only selection
OOT immutability	single final evaluation

Random forest wins narrowly; logistic regression remains available

Candidate	Val PR-AUC	Val ROC-AUC	Val Brier	Selection score	Status
Random forest	0.3768	0.7646	0.1099	0.4057	Champion
Logistic regression	0.3771	0.7622	0.1108	0.4053	Challenger

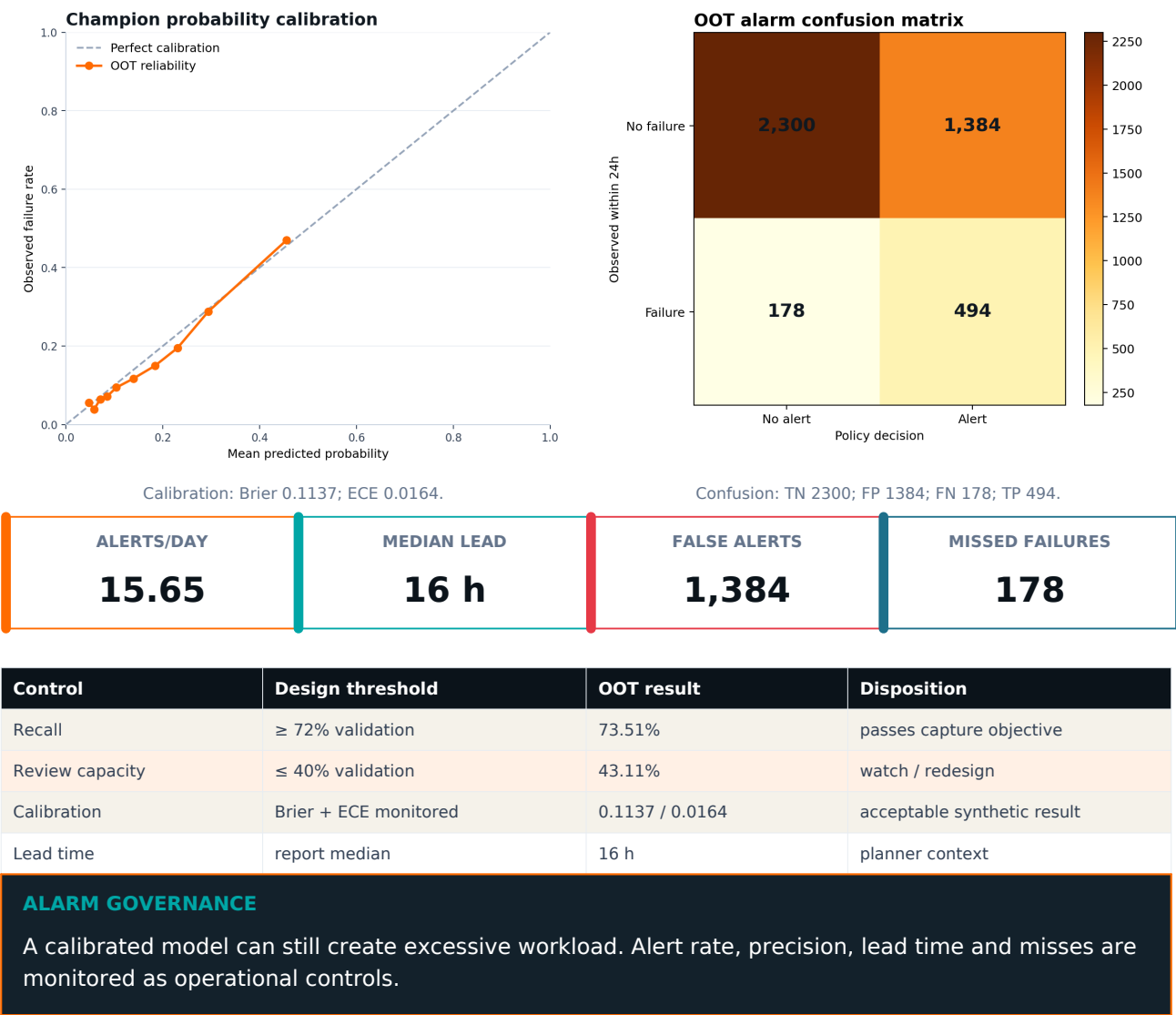
OOT PR-AUC 0.3887	OOT ROC-AUC 0.7489	OOT BRIER 0.1137	OOT ECE 0.0164
PRECISION 26.30%	RECALL 73.51%	F1 38.75%	ALERT RATE 43.11%

Measure	Validation	OOT	Interpretation
PR-AUC	0.3768	0.3887	ranking above baseline
Recall	75.78%	73.51%	failure capture
Precision	28.67%	26.30%	true failures among alerts
Alert rate	39.51%	43.11%	planner workload
Expected cost / obs.	416.5	469.9	synthetic scenario

VALIDATION CONCLUSION

The champion generalizes directionally, but OOT workload exceeds the validation capacity cap. Promotion requires operational review rather than metric-only approval.

Probability quality and planner workload are evaluated separately



Global drivers become local planner reason codes



Figure 7. Mean absolute SHAP value across 700 explained rows.

Rank	Reason code	SHAP value	Transformed value
1	TOOL_WEAR_ELEVATED	0.0773	2.021
2	LOW_LUBRICATION_INDEX	0.0619	-1.999
3	SUSTAINED_VIBRATION	0.0551	2.224
4	MAINTENANCE_INTERVAL_EXCEEDED	0.0535	2.427

LOCAL EXAMPLE

SH-MC-09-2025052508 • risk 63.62% • top reasons: TOOL_WEAR_ELEVATED, LOW_LUBRICATION_INDEX, SUSTAINED_VIBRATION, MAINTENANCE_INTERVAL_EXCEEDED.

- Global importance supports model review; local SHAP supports a specific planner conversation.
- Reason codes are deterministic labels layered on model features; they are not verified root causes.
- Correlated features can share or exchange attribution.
- SHAP explains the fitted base estimator; the sigmoid calibration layer is outside attribution.

Risk, criticality and synthetic economics determine review priority

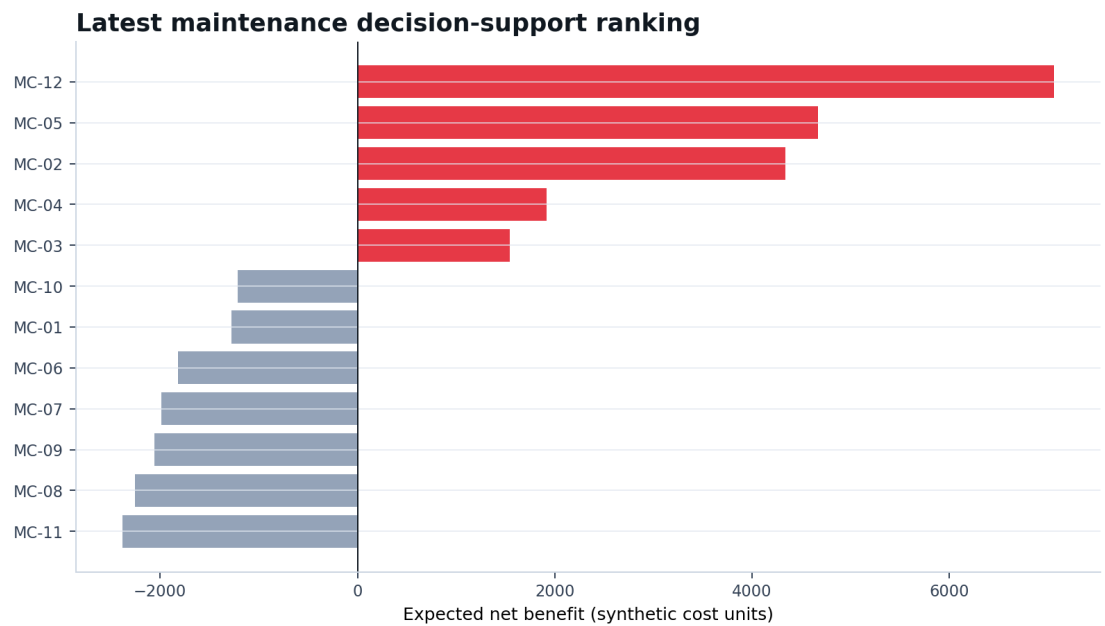


Figure 8. Latest OOT machine priority by expected net benefit.

Input	Role
Calibrated failure probability	likelihood within 24-hour horizon
Machine criticality	consequence multiplier tier 1-5
Estimated failure cost	synthetic scenario loss
Maintenance cost	synthetic intervention cost
Intervention effectiveness	scenario avoided-loss fraction

P1 REVIEWS	MONITOR	EXECUTION	HUMAN GATE
5	7	NONE	TRUE

Priority	Meaning	Required action
P1	high risk/criticality with positive expected net benefit	planner review within 8 hours
P2	above threshold with positive net benefit	planner review within 24 hours
P3	moderate risk or cost ratio	inspect in routine planning
MONITOR	no favorable recommendation	continue standard monitoring

SAFETY INVARIANT

The policy cannot override lockout/tagout, operating procedures, current equipment state or qualified maintenance judgment.

Every delivery channel consumes the same governed evidence

Layer	Deliverable	Control
FastAPI	recommendation-only endpoint, health and Prometheus metrics	human gate in response contract
PostgreSQL	normalized schema and analytical views	typed keys, timestamps and decision fields
Power BI PBIP	starter semantic model, DAX measures and six report pages	explicit project-start limitation
Excel	14-sheet formula workbook with dashboard and planner	blue inputs, comments and formula scan
PPT/PDF	executive narrative and governance evidence	results loaded from pipeline artifacts

API BOUNDARY

POST /v1/recommendations returns a probability, threshold, priority, reason codes and human approval fields. No write client for plant or maintenance systems exists.

ANALYTICAL DATABASE VIEWS

Governed consumption

View	Purpose
manufacturing.vw_oe_daily	daily OEE component and plant rollup
manufacturing.vw_downtime_pareto	duration/event Pareto by category
manufacturing.vw_machine_reliability	MTBF, MTTR and machine reliability
manufacturing.vw_quality_copq	FPY, scrap, rework and COPQ
manufacturing.vw_maintenance_priority	latest human-gated policy output

- PBIP is a version-controlled starter; it requires validation in the current Power BI Desktop.
- The Excel workbook is scenario-oriented and does not replace the pipeline system of record.
- All business-facing artifacts preserve the synthetic-data disclaimer.

Controls reduce portfolio risk without claiming production hardening

Threat	Example	Implemented / documented control
Input manipulation	extreme valid sensor payload	schema bounds, model drift and gateway guidance
Model substitution	modified joblib bundle	read-only mount, image immutability and hash runbook
Dependency compromise	vulnerable package/action	pins, pip-audit, Dependabot, CodeQL and Trivy
Credential leakage	secret committed to repository	.env example, external secret guidance and scanning
Denial of service	high request volume	resource limits, HPA and rate-limit guidance
Recommendation escalation	advice treated as work order	human gate and no CMMS/PLC client
Monitoring evasion	missing/forged metrics	independent scrape and missing-target alert
Data poisoning	shifted source distribution	contract checks, PSI and manual retraining gate

UNIT/COVERAGE 35 / 95.3%	PIP-AUDIT CLEAN	CODEQL WORKFLOW	TRIVY WORKFLOW
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Secure delivery control	Evidence
Static analysis	.github/workflows/codeql.yml
Dependency audit	.github/workflows/security.yml and local pip-audit
Container scan	Trivy filesystem and image jobs
Ownership	.github/CODEOWNERS
Dependency updates	.github/dependabot.yml
Protected main	docs/branch_protection_guide.md

OUT OF SCOPE

Plant segmentation, IAM integration, enterprise secrets management, signed telemetry and production CMMS connectivity remain deployment responsibilities.

Data quality, drift, model quality and workload have separate controls

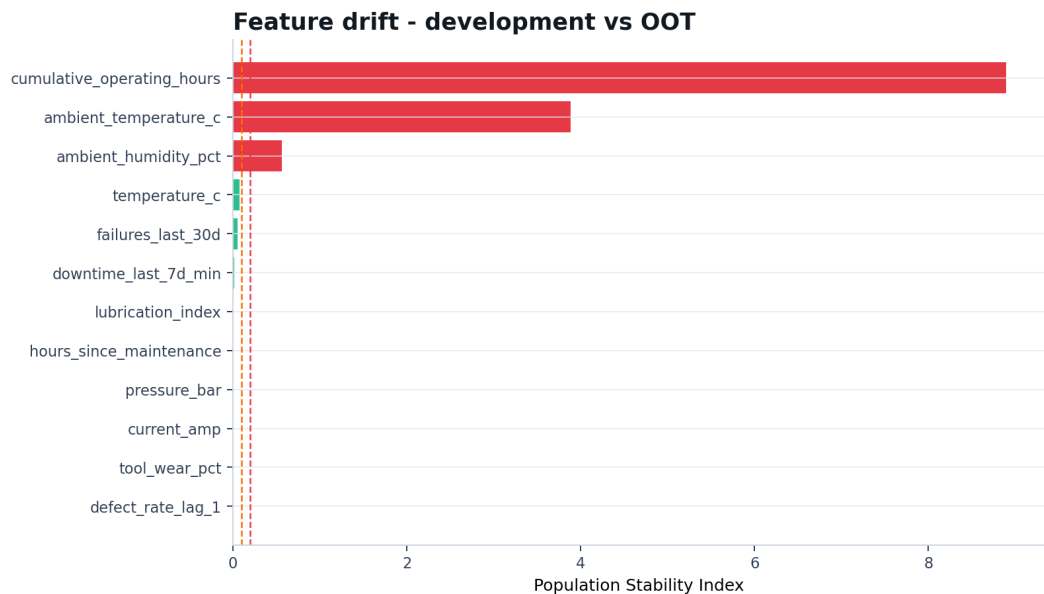


Figure 9. PSI comparing model-fit and OOT feature populations.

DATA CHECKS	ACTION PSI	MAX PSI	OOT ALERT RATE
5/5	3	8.899	43.11%

Signal class	Examples	First response
Data quality	missing/invalid/range violations	quarantine batch; verify source contract
Data drift	PSI action level	segment by time/source; do not auto-retrain
Model quality	PR-AUC, Brier, ECE, recall	compare to validation and business tolerance
Alarm quality	precision, alert rate, lead time, misses	review threshold and planner capacity
Service health	latency, errors, missing scrape	fail safe; investigate API/dependency

INCIDENT SEQUENCE

Detect → classify → contain → preserve evidence → assess decision impact → approve recovery → retrospective. Prediction/recommendation outages fail closed to manual planning.

Residual risk remains high for synthetic-to-real transfer

ID	Risk	Likelihood	Impact	Residual
R-01	Synthetic-to-real transfer gap	High	High	High
R-02	Temporal leakage	Medium	High	Low
R-03	Poor probability calibration	Medium	High	Medium
R-04	Alert overload	High	Medium	Medium
R-05	Seasonal/source drift	High	Medium	Medium
R-06	Control/specification confusion	Medium	High	Low
R-07	SHAP over-interpretation	Medium	Medium	Medium
R-08	Cost assumption sensitivity	High	Medium	Medium
R-09	Unauthorized operational integration	Low	Critical	Low
R-10	Dependency/container vulnerability	Medium	High	Medium
R-11	Model bundle tampering	Low	High	Medium
R-12	Missing/malformed sensor data	Medium	Medium	Medium

PRIMARY LIMITATION

Synthetic performance cannot establish real-plant effectiveness. Real adoption requires shadow data, verified labels, plant-specific cost and safety owners, and a new untouched acceptance window.

KNOWN VALIDATION LIMITATIONS

Items that block overclaiming

- OOT alert rate 43.11% exceeds the 40% validation review-capacity cap.
- Time/cumulative variables generate large PSI across a later window and require redesign review.
- Failure labels are synthetic and cleaner than real maintenance histories.
- Synthetic cost units are useful for ranking sensitivity, not finance approval.
- No causal inference or counterfactual maintenance-effect estimate is claimed.

Official sources are design references—not compliance claims

Reference	Concept	Project evidence	Claim boundary
ISO 22400-2	Manufacturing KPI formulas	OEE documentation	No certification/conformance
ISO 13374-2	Condition-monitoring processing	Staged architecture	Illustrative only
ISO 13374-4	Health/advisory presentation	Dashboard/recommendation contract	No presentation conformance
NIST SPC	Variables/attributes charts	X-bar/R, I-MR, p and u	Synthetic process
NIST capability	Capability/specification	Cp/Cpk/Pp/Ppk	No process acceptance
NIST AI RMF 1.0	Govern/map/measure/manage	Model card, monitoring and risks	Voluntary reference
NIST SSDF 1.1	Secure development	CI/tests/security scans	No SSDF audit
GitHub CodeQL	Code scanning	codeql.yml	Execution/config dependent
GitHub dependency review	Supply-chain changes	pip-audit/Dependabot	No complete risk guarantee
Prometheus rules	Operational alerts	rules + runbook	Thresholds need tuning

Official URL	Use
https://www.iso.org/standard/54497.html	ISO 22400-2
https://www.itl.nist.gov/div898/handbook/pmc/section3/pmc3.htm	SPC taxonomy
https://www.itl.nist.gov/div898/handbook/pmc/section1/pmc16.htm	Capability
https://www.nist.gov/itl/ai-risk-management-framework	AI risk
https://csrc.nist.gov/pubs/sp/800/218/final	SSDF
https://prometheus.io/docs/prometheus/latest/configuration/alerting_rules/	Alert rules

CROSSWALK STATUS

Versions and status must be rechecked before production use. The crosswalk documents design intent; it does not establish compliance.

One evidence chain across code, analytics and communication

Delivery	Location	Review question
Python pipeline	src/manufacturing_ct/	Can the result be reproduced?
Tests and QA	tests/ and .github/workflows/	Are controls automated and covered?
SQL	sql/	Can analytical records be governed and queried?
Power BI PBIP	powerbi/	Can decision consumers explore the model?
Excel	deliverables/excel/	Can assumptions and priorities be reviewed?
Executive deck	deliverables/presentation/	Can leaders understand the decision?
Governance PDF	deliverables/report/	Can reviewers trace claims to evidence?
Documentation	docs/	Are contracts, ADRs and runbooks explicit?
Portfolio copy	docs/portfolio/	Are public claims bounded and accurate?
Infrastructure	Dockerfile, compose, k8s/, monitoring/	Can operability be inspected?

PIPELINE ROWS	TESTS	COVERAGE	HUMAN GATE
19,692	35	95.30%	ENFORCED

FINAL GOVERNANCE POSITION

The project demonstrates an end-to-end manufacturing analytics and predictive-maintenance decision system while preserving synthetic-data, human-approval and non-compliance boundaries.

PROPOSED NEXT STEP

If real data becomes available

- Run in shadow mode with plant-specific data contracts and verified failure labels.
- Rebuild temporal splits, recalibrate synthetic economics and define planner capacity.
- Conduct safety, cybersecurity, privacy and architecture reviews with accountable owners.
- Approve any integration only after monitored acceptance criteria and rollback plans.